

Drug -> Target -> Pathway -> Transcription -> Synergy

An expanded biology and ML guide for drug-synergy research

Core idea: A drug usually does not directly affect a gene. It usually binds one or more proteins, changes protein activity, propagates through pathways and networks, changes transcription-factor activity, and then creates a gene-expression response. Your ML architecture should respect those biological layers.

Guide map

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3	mRNA abundance, protein abundance, enzyme activity	Biological layers and factory analogy.
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5	How drugs are related to genes	Drug -> target -> pathway -> TF -> expression.
6	Small-molecule binding, targets, off-targets, potency, MoA	Binding, IC50, EC50, off-target effects, mechanism.
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9	JAK/STAT, p53, NF-kB	Cytokine signaling, stress response, inflammation/survival.
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1. DNA -> RNA -> protein: the central dogma

The central dogma is the basic information flow of molecular biology. DNA is the stable instruction store, RNA is a copied or working message, and proteins are the functional molecules that carry out most cellular work.



Figure 1. Central information flow from stored genetic information to cellular function.

Restaurant analogy: DNA is the master cookbook kept safe in the office. mRNA is a temporary photocopy of one recipe. The ribosome is the chef reading the recipe. The protein is the finished dish. The cell phenotype is what the dish does to the restaurant service.

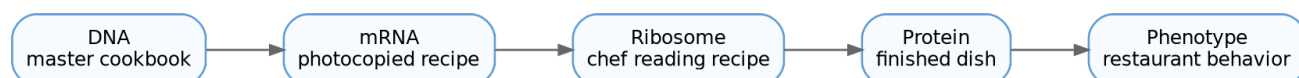


Figure 2. Restaurant analogy for gene expression.

Research consequence: RNA-seq mostly measures copied recipes. It does not directly measure how many finished protein dishes were made or whether those proteins are active.

Abbreviations in this section

Abbreviation	Expansion	Meaning
DNA	Deoxyribonucleic acid	Stable hereditary material.
RNA	Ribonucleic acid	A family of molecules involved in expression and regulation.
mRNA	Messenger RNA	The protein-coding RNA copy of a gene.
RNA-seq	RNA sequencing	A method for measuring RNA abundance across many genes.

2. RNA vs mRNA

RNA is not one thing. It is a family of related molecules. mRNA is the subtype that carries protein-coding instructions, but other RNAs help with translation, ribosome structure, and regulation.

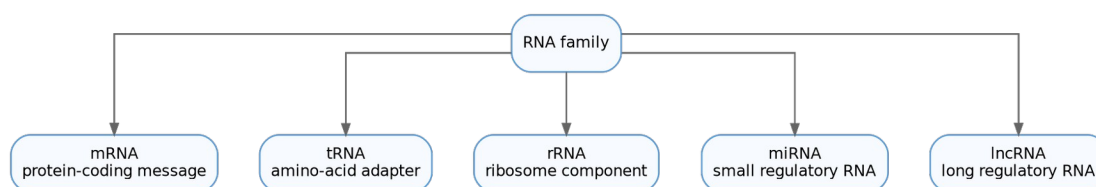


Figure 3. All mRNA is RNA, but not all RNA is mRNA.

RNA type	Role	Why it matters
mRNA	Carries protein-coding information.	Most RNA-seq gene-expression matrices focus on this layer.
tRNA	Brings amino acids during translation.	Required to build proteins from mRNA.
rRNA	Structural/catalytic component of ribosomes.	Makes the translation machine possible.
miRNA	Regulates target transcripts.	Can reduce translation or destabilize mRNA.
lncRNA	Regulatory or scaffolding functions.	Can affect chromatin, transcription, or

protein complexes.

Abbreviations in this section

Abbreviation	Expansion	Meaning
tRNA	Transfer RNA	Adapter RNA used in translation.
rRNA	Ribosomal RNA	RNA component of ribosomes.
miRNA	MicroRNA	Small RNA involved in post-transcriptional regulation.
lncRNA	Long non-coding RNA	Long RNA molecule that usually does not code for protein.

3. mRNA abundance, protein abundance, and enzyme activity

These are different biological layers. mRNA abundance is message level. Protein abundance is molecule level. Enzyme or protein activity is functional level. A model that treats them as equivalent will miss important biology.

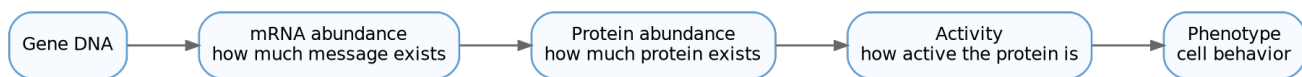


Figure 4. Biological layers from gene to phenotype.

Layer	Factory analogy	Typical measurement	Failure mode
mRNA abundance	Number of printed work orders.	RNA-seq or qPCR.	Many orders do not guarantee many machines.
Protein abundance	Number of machines built.	Proteomics or Western blot.	Many machines do not guarantee they are on.
Enzyme activity	Whether machines are powered and producing output.	Activity assay or phospho-protein assay.	Activity can change without expression change.

ML consequence: Basal expression can help describe cell state, but it is not a complete measurement of pathway activity. Use pathway and regulon scoring to get closer to functional state.

Abbreviations in this section

Abbreviation	Expansion	Meaning
qPCR	Quantitative polymerase chain reaction	A targeted method for measuring DNA/RNA abundance.
Western blot	Protein immunoblot	A method for detecting specific proteins.

4. Why expression does not equal activity

Gene expression usually means mRNA level. But proteins are regulated after transcription by translation efficiency, degradation, chemical modification, localization, binding partners, and inhibitors.

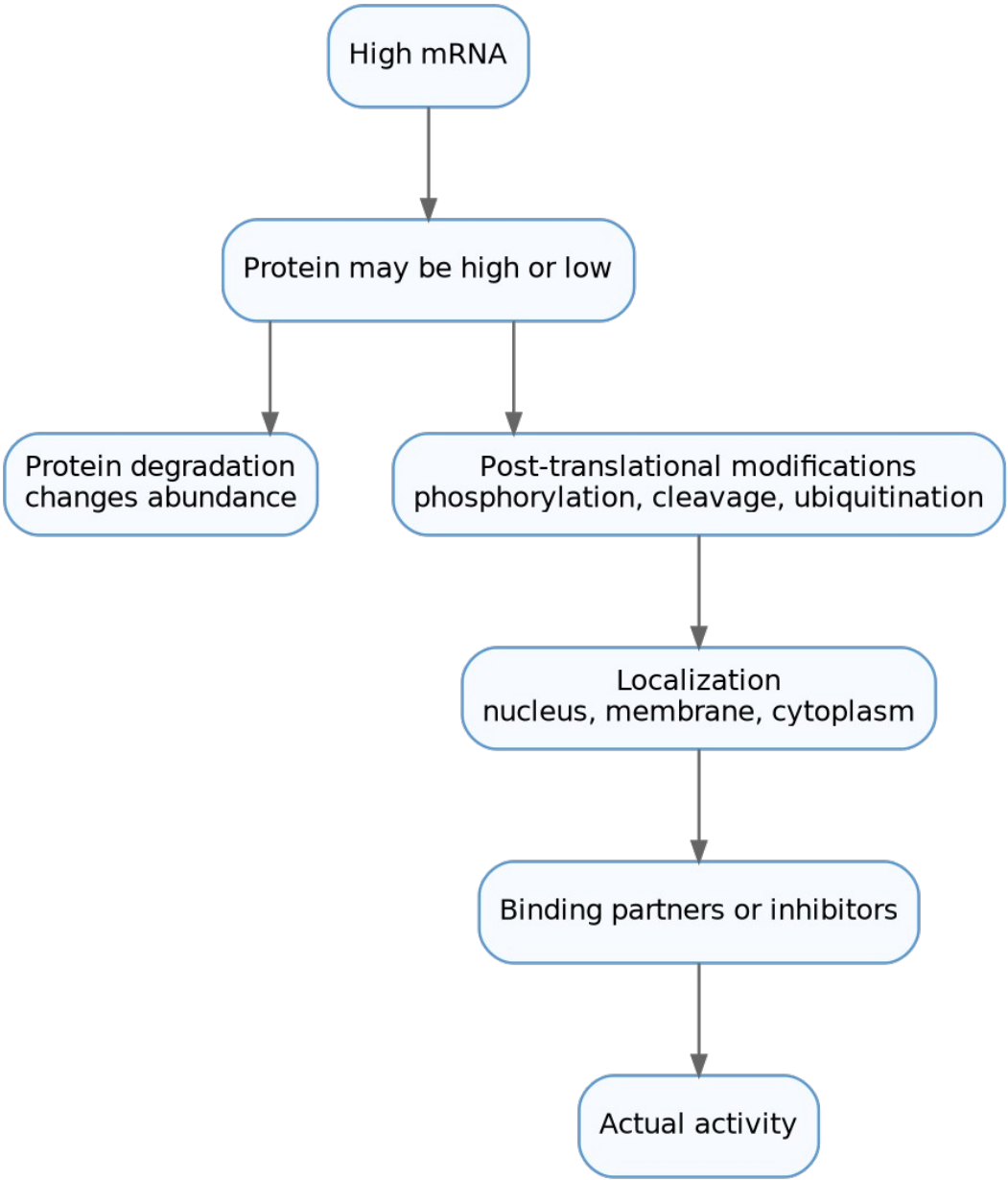


Figure 5. Regulatory gates between mRNA and activity.

Example: AKT transcript may be abundant, but AKT pathway activity is often better reflected by phosphorylated AKT and downstream responsive genes. Similarly, a transcription factor may be present but inactive until it enters the nucleus.

Key principle: Expression is a useful proxy, not ground truth. Protein activity is often controlled by state, not quantity.

Abbreviations in this section

Abbreviation	Expansion	Meaning
PTM	Post-translational modification	A modification made to a protein after translation.
AKT	Protein kinase B	A kinase in the PI3K/AKT pathway.
Phosphorylation	Addition of a phosphate group	A common switch for activating or inhibiting proteins.

5. How drugs are related to genes

A drug often appears to affect many genes, but its direct molecular event is usually protein binding. The gene-expression effect is downstream.



Figure 6. Most drug-to-gene effects are indirect.

Better wording: instead of saying “drug X affects gene Y,” say “drug X binds target T, changes pathway P, changes transcription factor activity, and this changes expression of gene Y.”



Figure 7. A MEK inhibitor can change hundreds of downstream genes without binding those genes directly.

Abbreviations in this section

Abbreviation	Expansion	Meaning
MEK	MAPK/ERK kinase	A kinase in the MAPK pathway.
ERK	Extracellular signal-regulated kinase	A downstream kinase in the MAPK pathway.
ELK1	ETS-like transcription factor 1	A transcription factor regulated by MAPK signaling.
AP-1	Activator protein 1	A transcription factor complex often downstream of MAPK/JNK signaling.

6. Small-molecule binding, targets, off-targets, potency, and MoA

A small molecule drug is a chemical compound that can bind proteins. The intended protein is the target. Other proteins bound unintentionally are off-targets. Off-targets can cause toxicity, resistance, or unexpected synergy.



Figure 8. Direct drug action usually starts with binding.

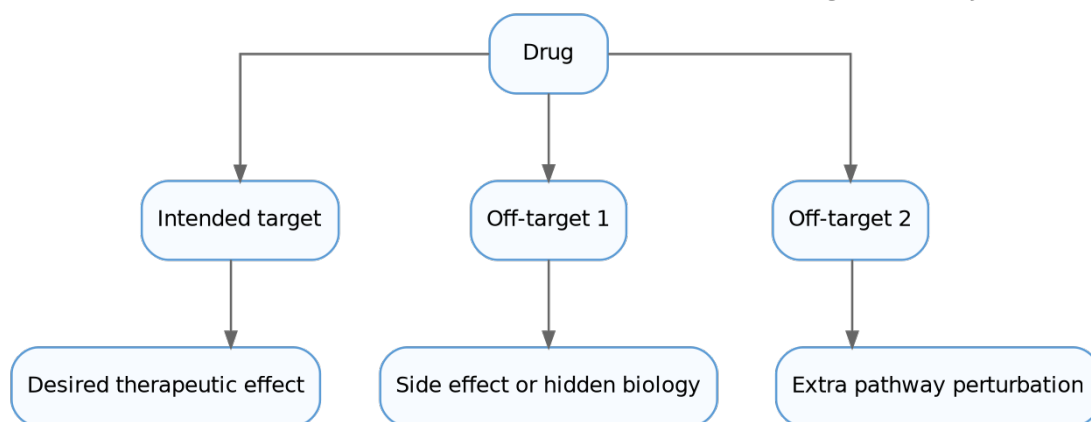


Figure 9. A drug can perturb more than its intended target.

Term	Meaning	Synergy relevance
IC50	Dose causing 50% inhibition in a specific assay.	Lower IC50 can mean stronger target inhibition at lower concentration.
EC50	Dose causing 50% of maximal effect.	Useful for activators or phenotypic effect curves.
MoA	The chain of molecular events producing the effect.	MoA similarity or complementarity can explain drug-pair behavior.
Off-target	Unintended target interaction.	Can create unexpected pair effects.

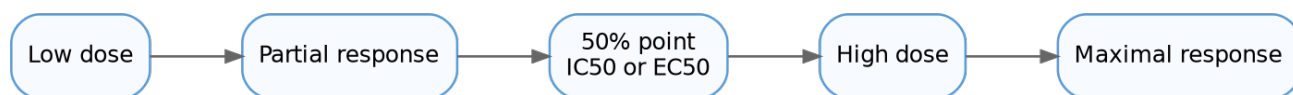


Figure 10. Conceptual dose-response curve simplified as stages.

Abbreviations in this section

Abbreviation	Expansion	Meaning
IC50	Half-maximal inhibitory concentration	Concentration causing 50% inhibition.
EC50	Half-maximal effective concentration	Concentration producing 50% of maximal effect.
MoA	Mechanism of action	How a drug causes its biological effect.

7. Signaling pathway master flow

Signaling pathways convert signals into decisions. A receptor or sensor detects a signal, kinase/adaptor proteins relay it, transcription factors implement regulatory changes, and the cell phenotype changes.

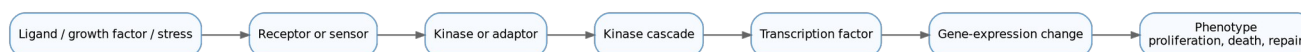


Figure 11. General receptor-to-transcription signaling logic.

Company analogy: a signal is an external email, the receptor is reception, kinase cascades are managers forwarding instructions, transcription factors are executives changing policy, and gene expression is the new instruction set distributed to employees.

Modeling consequence: Pathways are intermediate representations between direct targets and measured expression. They help your model avoid treating every gene independently.

Abbreviations in this section

Abbreviation	Expansion	Meaning
Kinase	Protein kinase	An enzyme that modifies proteins by phosphorylation.
TF	Transcription factor	A protein that regulates transcription.

8. Major signaling pathways I: MAPK and PI3K/AKT

8.1 MAPK pathway

MAPK signaling is strongly associated with growth and proliferation. In many cancers, receptor activation or mutations keep this pathway on.



Figure 12. MAPK signaling from growth-factor detection to proliferation genes.

8.2 PI3K/AKT pathway

PI3K/AKT signaling supports survival, growth, metabolism, and resistance. It is often a parallel survival route when growth signaling is blocked.

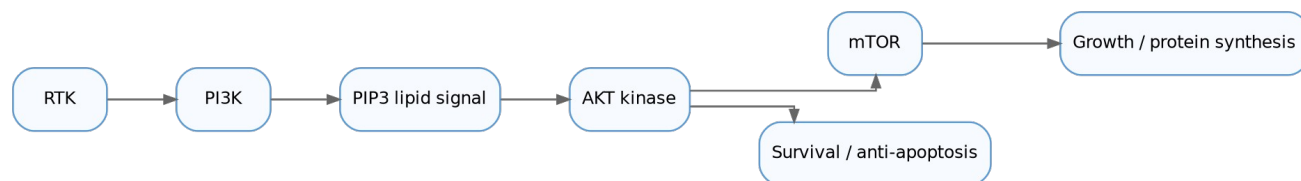


Figure 13. PI3K/AKT signaling supports survival and growth.

Synergy example: A MEK inhibitor can reduce proliferation signaling, while a PI3K or AKT inhibitor can reduce survival signaling. In the right cell context, blocking both can produce stronger killing than either alone.

Abbreviations in this section

Abbreviation	Expansion	Meaning
MAPK	Mitogen-activated protein kinase	Pathway family that relays growth or stress signals.
RTK	Receptor tyrosine kinase	A receptor that signals through tyrosine phosphorylation.
RAS	Rat sarcoma small GTPase family	A molecular switch upstream of RAF.
RAF	Rapidly accelerated fibrosarcoma kinase	A kinase downstream of RAS.
MEK	MAPK/ERK kinase	A kinase that activates ERK.
ERK	Extracellular signal-regulated kinase	A downstream MAPK.
PI3K	Phosphoinositide 3-kinase	Creates lipid signals for AKT activation.
PIP3	Phosphatidylinositol (3,4,5)-trisphosphate	A membrane lipid signal.
mTOR	Mechanistic target of rapamycin	A growth and protein-synthesis regulator.

9. Major signaling pathways II: JAK/STAT, p53, and NF- κ B

9.1 JAK/STAT

JAK/STAT signaling converts cytokine signals into transcriptional programs. It is important in immune signaling, inflammation, proliferation, and some cancer contexts.

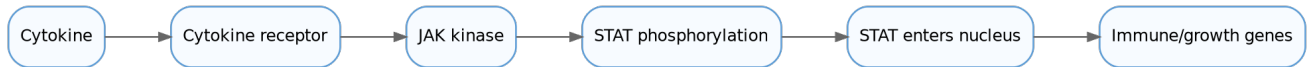


Figure 14. JAK/STAT signaling.

9.2 p53

p53 is a stress-response tumor suppressor. It can drive cell-cycle arrest, DNA repair, senescence, or apoptosis depending on damage severity and cellular context.

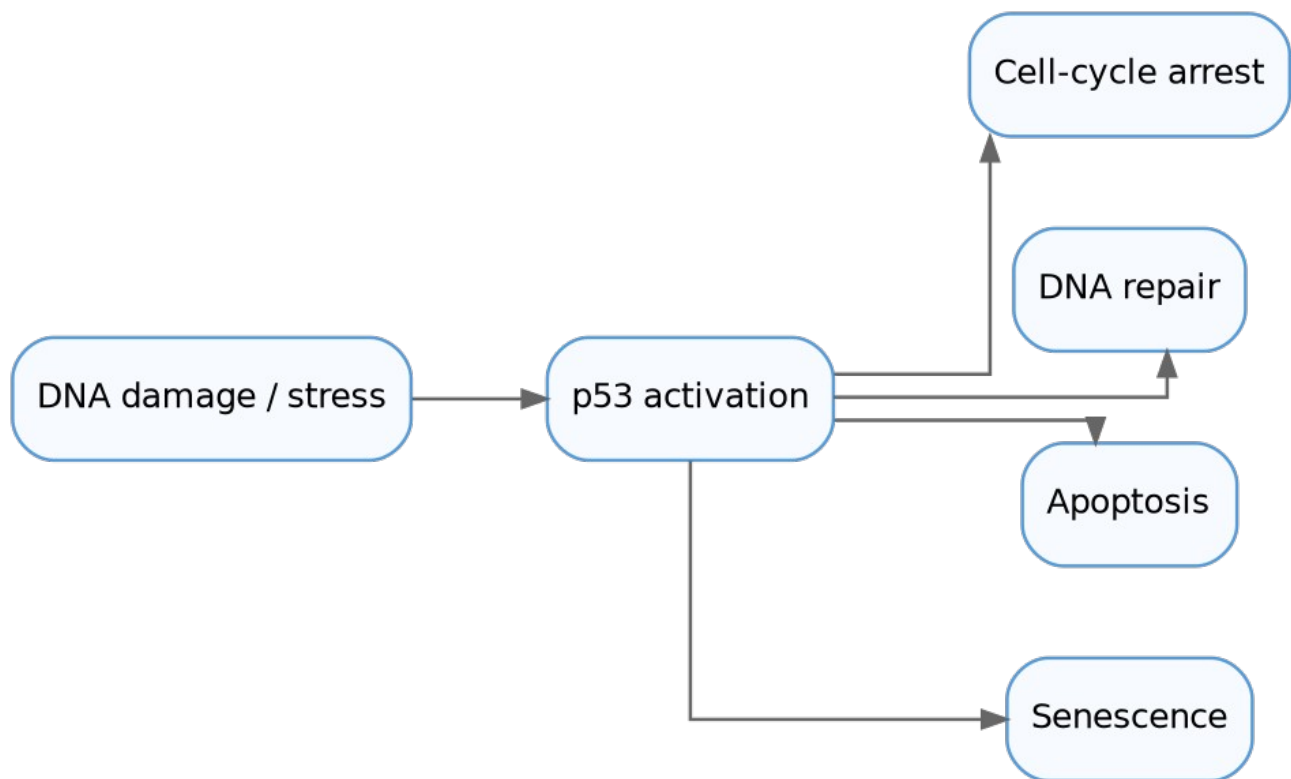


Figure 15. p53 response decisions.

9.3 NF- κ B

NF- κ B coordinates inflammatory, immune, stress, and survival programs. It can protect cells from apoptosis and contribute to treatment resistance.



Figure 16. NF- κ B activation logic.

Abbreviations in this section

Abbreviation	Expansion	Meaning
JAK	Janus kinase	A kinase downstream of cytokine receptors.
STAT	Signal transducer and activator of transcription	A transcription factor activated by JAK signaling.
p53	Tumor protein p53	A tumor-suppressor transcription factor.
NF-kB	Nuclear factor kappa-light-chain-enhancer of activated B cells	A transcription-factor family involved in inflammation and survival.
TNF	Tumor necrosis factor	A cytokine that can activate inflammatory and death/survival signaling.
IKK	I-kappa-B kinase	A kinase complex that activates NF-kB.
IkB	Inhibitor of kappa B	An inhibitor that retains NF-kB before activation.

10. Major cellular programs: cell cycle, apoptosis, and DNA damage response

10.1 Cell cycle

The cell cycle controls whether and when a cell divides. Checkpoints prevent division when DNA is damaged or replication is incomplete.

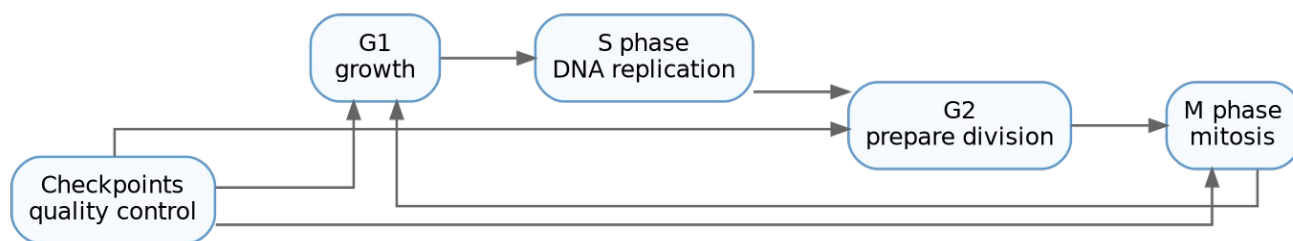


Figure 17. Cell-cycle phases and checkpoints.

10.2 Apoptosis

Apoptosis is programmed cell death. Many anticancer synergies work by lowering the threshold for apoptosis after blocking survival pathways or increasing stress.

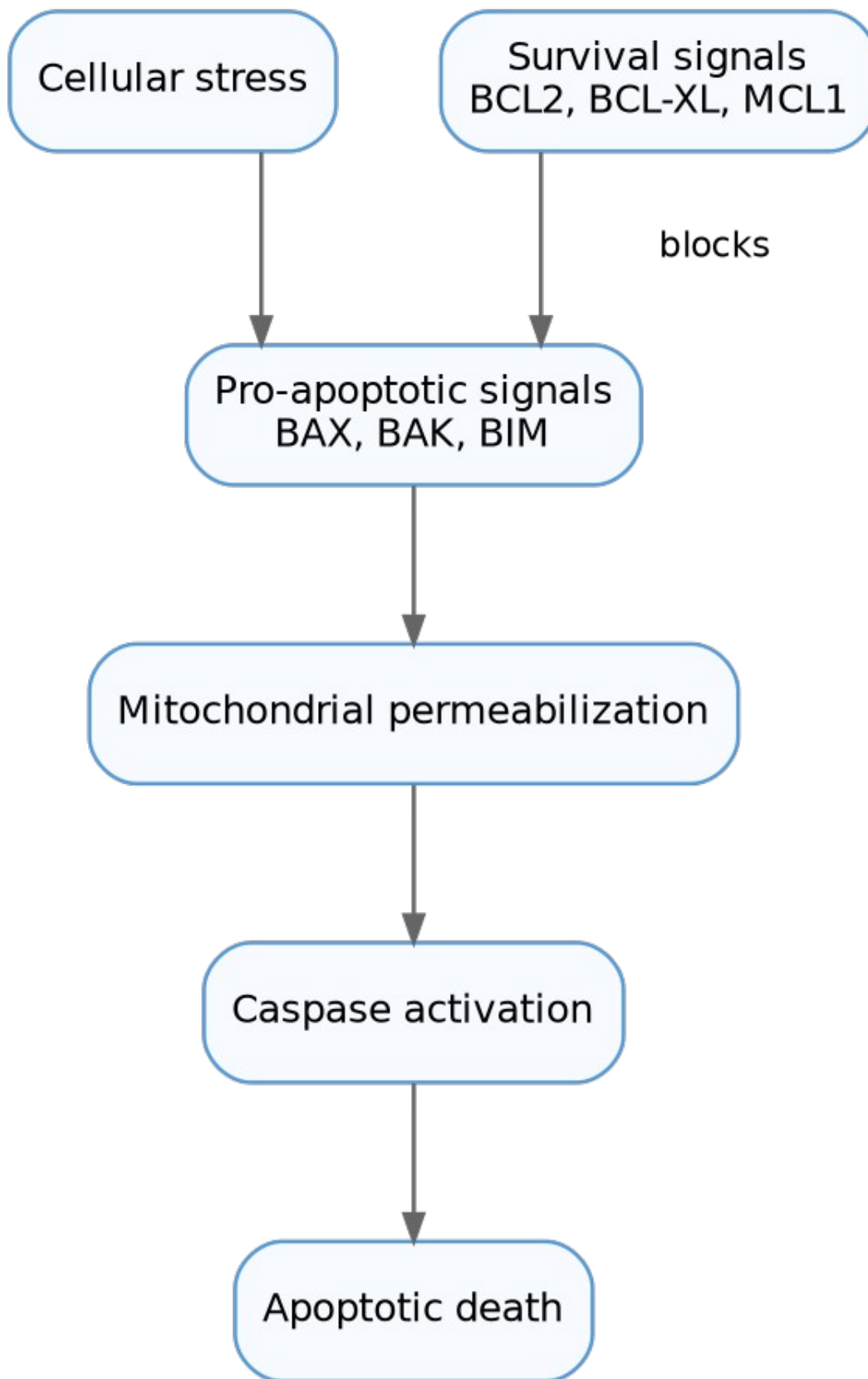


Figure 18. Simplified apoptosis logic.

10.3 DNA damage response

The DNA damage response detects DNA lesions, pauses the cell cycle, coordinates repair, or escalates to death if damage is too severe.

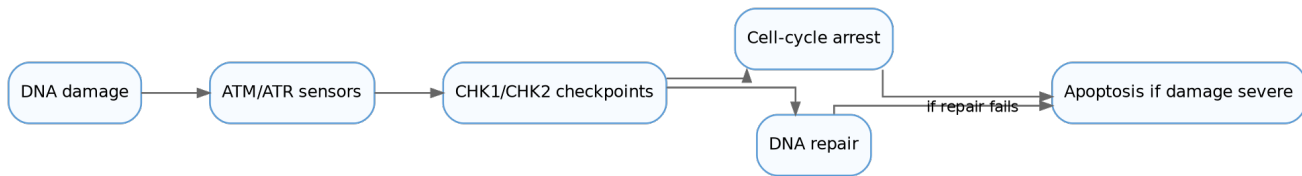


Figure 19. DNA damage response logic.

Abbreviations in this section

Abbreviation	Expansion	Meaning
G1/S/G2/M	Cell-cycle phases	Growth, DNA synthesis, preparation, and mitosis.
CDK	Cyclin-dependent kinase	A cell-cycle kinase family.
BAX/BAK/BIM	Pro-apoptotic BCL2-family proteins	Proteins that promote mitochondrial apoptosis.
BCL2/BCL-XL/MCL1	Anti-apoptotic BCL2-family proteins	Proteins that suppress apoptosis.
DDR	DNA damage response	Systems that detect and respond to DNA damage.
ATM/ATR	Ataxia-telangiectasia mutated / ATM and Rad3-related	DNA damage sensor kinases.
CHK1/CHK2	Checkpoint kinase 1/2	Checkpoint kinases activated by DNA damage.
PARP	Poly(ADP-ribose) polymerase	DNA repair enzyme family targeted by PARP inhibitors.

11. Transcription factors, TF-target genes, and regulons

A transcription factor is a protein that binds regulatory DNA and controls transcription. A TF-target gene edge means a TF regulates a gene. A regulon is the full set of genes regulated by a TF.

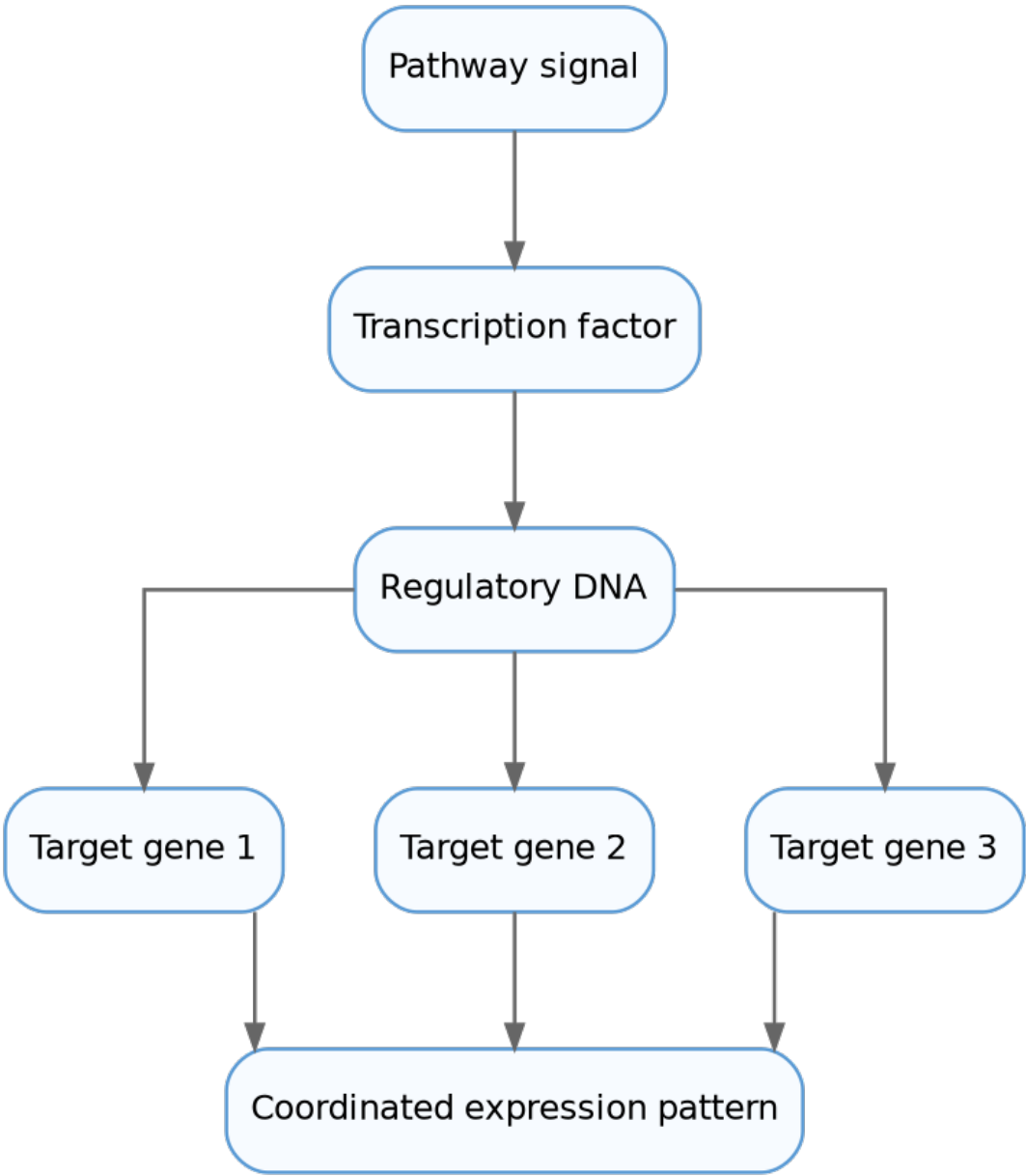


Figure 20. A regulon creates a coordinated target-gene pattern.

Regulation type	Expected target-gene pattern	Interpretation
Activation	Activated targets go up when TF activity rises.	The TF likely promotes transcription of those targets.
Repression	Repressed targets go down when TF activity rises.	The TF likely suppresses transcription of those targets.
Mixed signed regulon	Some targets are activated, others repressed.	Directionality improves activity inference.

CEO analogy: you do not infer CEO activity by measuring whether the CEO exists in the building. You infer it by observing coordinated changes in employee behavior. Likewise, TF activity is inferred from target genes, not just TF expression.

Abbreviations in this section

Abbreviation	Expansion	Meaning
TF	Transcription factor	DNA-binding regulator of transcription.

Regulon	Regulatory gene set	The collection of targets regulated by one regulator.
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12. TF activity inference: DoRothEA and VIPER concept

TF mRNA is often a poor proxy for TF activity. DoRothEA provides signed TF-target regulatory relationships, and VIPER-like methods use the expression behavior of those targets to infer regulator activity.

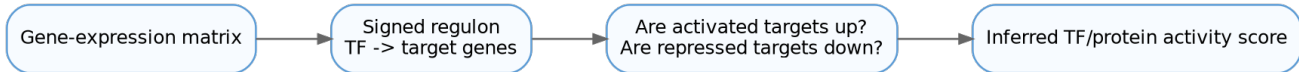


Figure 21. Regulon-based activity inference.

Encoder relevance: Instead of giving your model only thousands of gene values, you can give it inferred TF activities. This turns expression into regulatory state: MYC active, p53 suppressed, NF-kB high, and so on.

Abbreviations in this section

Abbreviation	Expansion	Meaning
DoRothEA	Discriminant Regulon Expression Analysis	A resource of signed TF-target interactions.
VIPER	Virtual Inference of Protein-activity by Enriched Regulon analysis	A method for inferring regulator activity from regulon expression.
MYC	MYC proto-oncogene transcription factor	A major growth and proliferation-associated TF.

13. Pathway perturbation becomes gene-expression change

A pathway perturbation becomes a gene-expression change through transcription factors and other regulators. This is why a drug targeting one protein can create a broad transcriptomic signature.



Figure 22. From direct drug target to observed gene-expression signature.

This also explains why basal expression and perturbation expression mean different things. Basal expression describes the pre-treatment state. Perturbation expression describes the response after drug, dose, time, and cell-line context are applied.

Important distinction: If your model uses only basal cell-line expression, it sees context. If it uses LINCS/L1000 perturbation signatures, it sees drug-induced response. These are complementary.

Abbreviations in this section

Abbreviation	Expansion	Meaning
Basal expression	Pre-treatment expression	Gene expression before perturbation.
Perturbation signature	Post-treatment expression response	Genes up/down after a drug or genetic perturbation.

14. Database edge types: do not mix incompatible biology

Different biological databases answer different questions. A drug-target edge, a protein-protein interaction edge, a pathway membership edge, and a TF-target edge are not the same type of evidence.

Database family	Examples	What the edge means	Good use
Pathway/reaction	KEGG, Reactome	Molecules participate in curated pathways, reactions, or events.	Map targets to biological processes.
Protein interaction/association	STRING, BioGRID	Proteins physically interact or are functionally associated.	Build target neighborhoods.
Transcriptional regulation	DoRothEA, TRRUST, ChEA	TF regulates target gene.	Infer TF/regulon activity.
Drug-target	DrugBank, ChEMBL, DGIdb	Drug binds or modulates a target.	Connect compounds to direct biology.
Perturbation/evidence	LINCS, L1000, CMap, CTD	Chemical or genetic perturbation changes expression or has evidence links.	Measure or support drug-induced response.

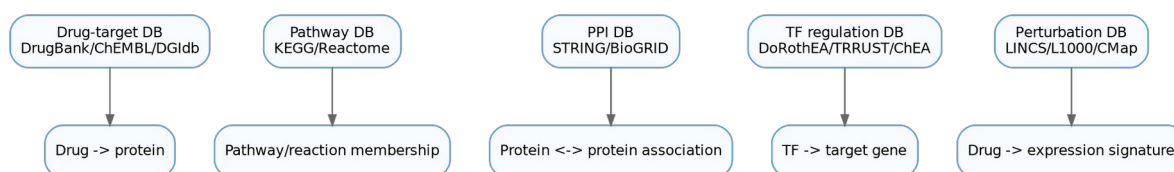


Figure 23. Database families encode different edge types.

Abbreviations in this section

Abbreviation	Expansion	Meaning
KEGG	Kyoto Encyclopedia of Genes and Genomes	Pathway and systems biology maps.
STRING	Search Tool for the Retrieval of Interacting Genes/Proteins	Protein association network resource.
BioGRID	Biological General Repository for Interaction Datasets	Biological interaction database.
TRRUST	Transcriptional Regulatory Relationships Unraveled by Sentence-based Text mining	TF-target regulation resource.
ChEA	ChIP-X Enrichment Analysis	Resource for TF-gene evidence from ChIP studies.
DrugBank	Drug and target knowledgebase	Drug, target, and pharmacology information.
ChEMBL	Chemical database of bioactive molecules	Bioactivity and target data.
DGIdb	Drug-Gene Interaction database	Drug-gene interaction search resource.
CTD	Comparative Toxicogenomics Database	Chemical-gene-disease evidence.

15. Gene set enrichment and pathway activity scoring

Raw expression can contain around 20,000 genes. Scoring methods compress that vector into interpretable biological programs: pathway activity, regulon activity, cell-cycle state, apoptosis state, and other modules.

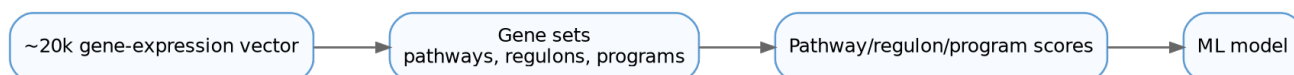


Figure 24. Converting gene-level expression into biological program scores.

Method	Core idea	Useful output
GSEA	Test whether genes from a gene set cluster near the top/bottom of a ranked list.	Enrichment of pathways between conditions.
ssGSEA	Single-sample version of GSEA-like scoring.	One pathway score per sample.
GSVA	Estimate pathway activity variation across	Sample-by-pathway matrix.

	samples from expression data.	
PROGENy	Infer pathway activity from perturbation-responsive footprint genes.	Pathway activity scores such as MAPK, PI3K, NF-kB.
VIPER	Infer regulator activity from regulon target-gene patterns.	TF/protein activity scores.

Abbreviations in this section

Abbreviation	Expansion	Meaning
GSEA	Gene Set Enrichment Analysis	A ranked-list gene-set enrichment method.
ssGSEA	Single-sample Gene Set Enrichment Analysis	A method that scores gene sets per sample.
GSVA	Gene Set Variation Analysis	A method for estimating pathway variation across samples.
PROGENy	Pathway RespOnsive GENes	A method inferring pathway activity from responsive genes.

16. Pathway activity score vs regulon activity score

Pathway scores and regulon scores are related but not identical. Pathway scores summarize biological process activity. Regulon scores summarize regulator or transcription factor activity. Both are useful features for interpreting cell state and drug response.

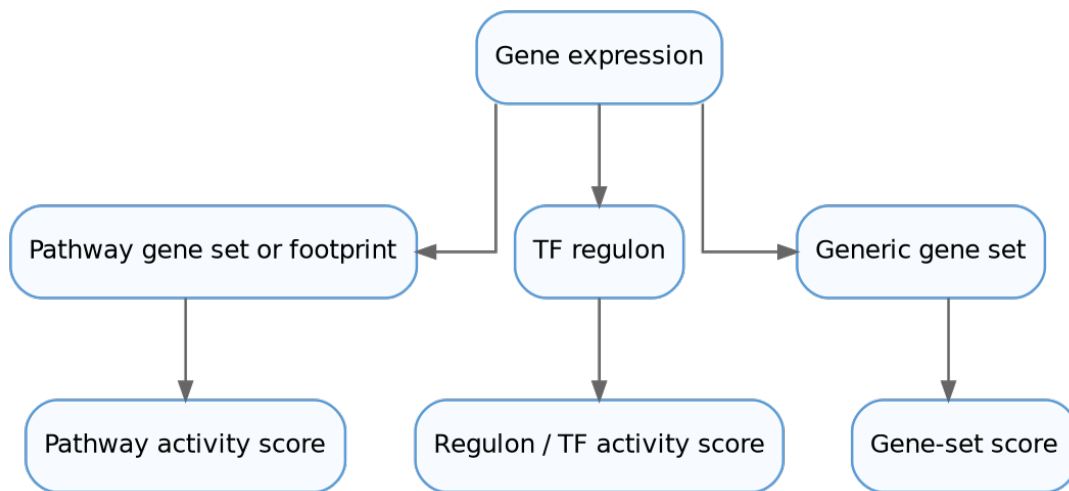


Figure 25. Different scoring frameworks extract different biological summaries.

Score type	Input prior knowledge	Output meaning	Example
Pathway activity	Pathway genes or perturbation footprint.	Activity of a signaling/process pathway.	MAPK high, PI3K high.
Regulon activity	TF-target gene network.	Activity of a regulatory protein/TF.	MYC active, p53 active.
Gene-set score	Any curated gene set.	Enrichment of a program.	Cell-cycle high, apoptosis low.

Abbreviations in this section

Abbreviation	Expansion	Meaning
Footprint	Downstream response pattern	Genes that change when a pathway/regulator is perturbed.

17. Network biology basics

Network biology represents biological systems as graphs. Nodes can be genes, proteins, drugs, or pathways. Edges can represent binding, regulation, interaction, activation, inhibition, or association.

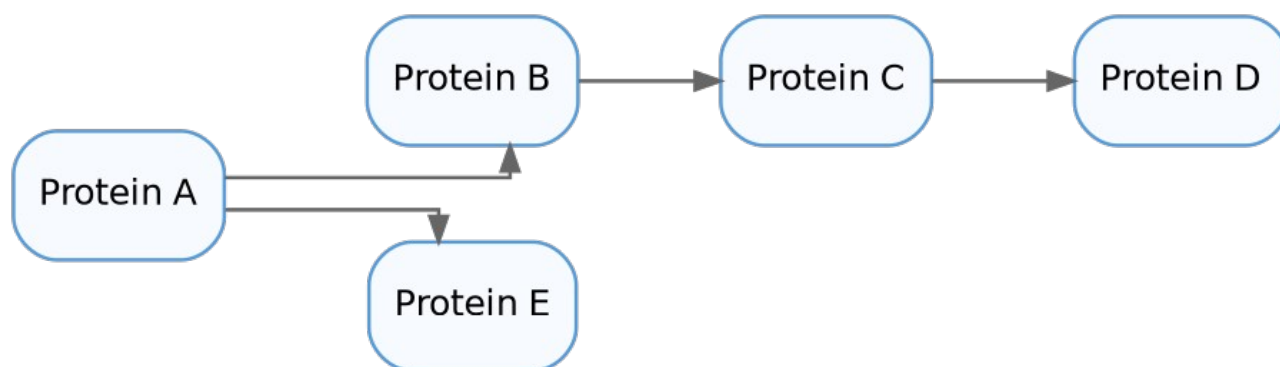


Figure 26. A small protein network.

Graph concept	Meaning in biology	Example
Node	Biological entity.	Gene, protein, drug, pathway.
Edge	Relationship between entities.	Binds, regulates, interacts, activates, inhibits.
Directed edge	Relationship has direction.	TF -> target gene.
Undirected edge	Association has no direction or direction is ignored.	Protein A -- Protein B.

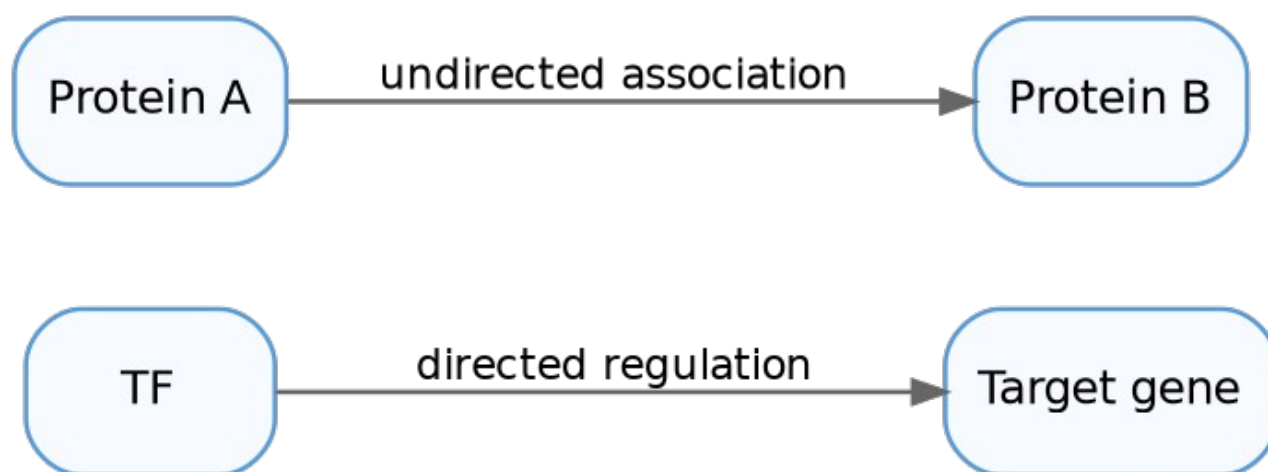


Figure 27. Directed and undirected edges mean different things.

Abbreviations in this section

Abbreviation	Expansion	Meaning
PPI	Protein-protein interaction	An interaction or association between proteins.

18. Graph neighborhoods, propagation, random walks, proximity, and modules

A drug target is not biologically isolated. Its graph neighborhood includes nearby proteins that may participate in the same pathway, complex, or functional process. Expanding from targets can produce richer drug-biology features.

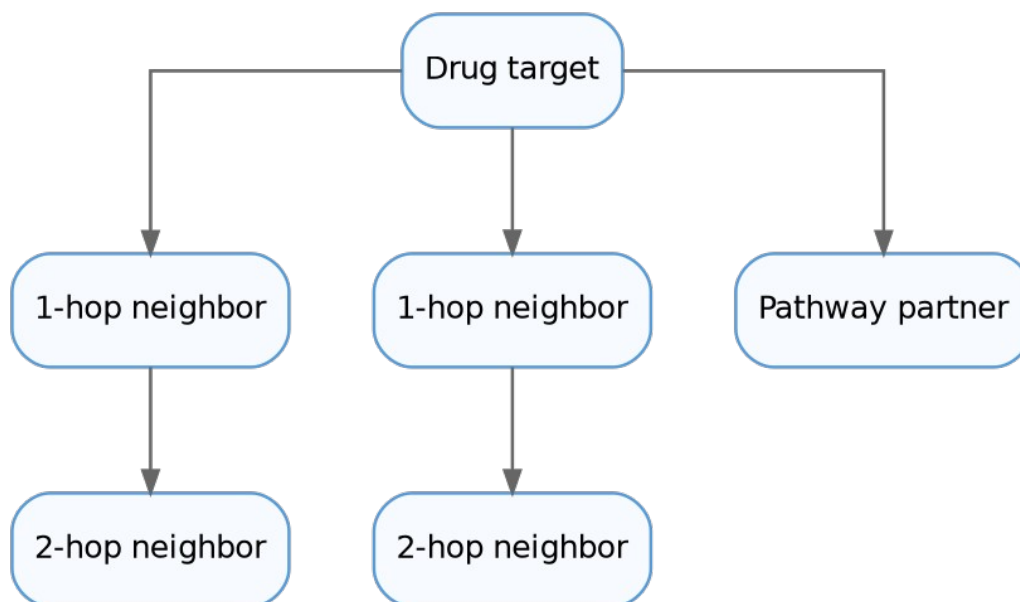


Figure 28. Target neighborhood expansion.

Network propagation spreads signal from seed nodes through the graph. Random walk with restart repeatedly walks to neighbors but sometimes jumps back to the seed targets, producing a smooth score over the network.

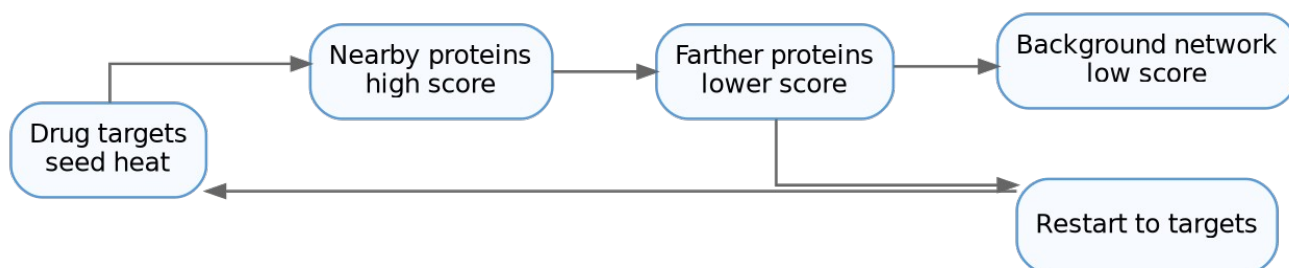


Figure 29. Network propagation and random-walk-with-restart intuition.

Network proximity asks whether two drugs target nearby biological regions. Module detection asks whether proteins cluster into functional units such as MAPK signaling, DNA repair, or apoptosis.

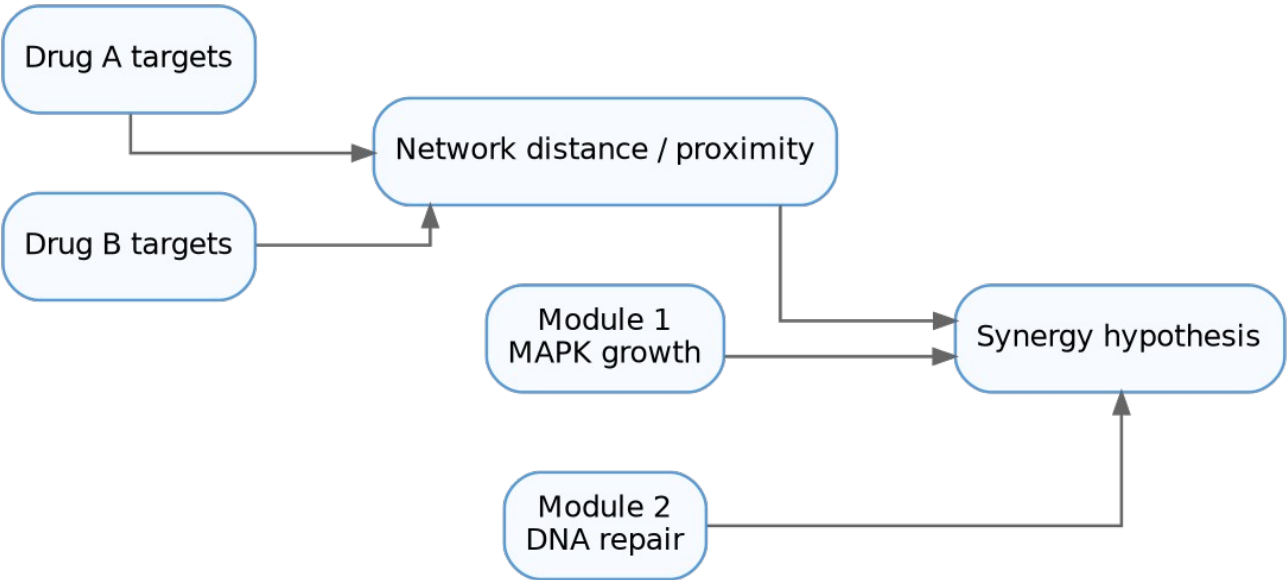


Figure 30. Network proximity and modules can create pair-level features.

Abbreviations in this section

Abbreviation	Expansion	Meaning
RWR	Random walk with restart	Graph algorithm that repeatedly returns to seed nodes.
Seed node	Starting node	Initial target or gene set for propagation.
Module	Network cluster	A group of connected nodes with related function.

19. Perturbational transcriptomics: LINCS L1000 and CMap

Perturbational transcriptomics measures gene expression after perturbing the cell. For “drug affects gene,” this is often the most direct evidence, because it observes expression after treatment.

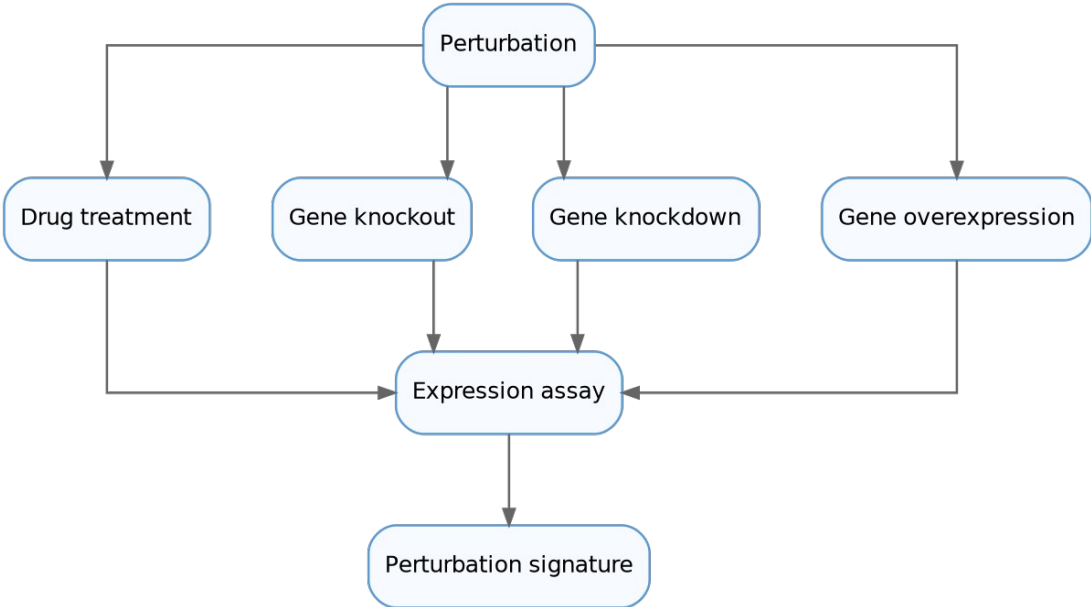


Figure 31. Perturbational transcriptomics measures response after an intervention.

Signature component	Meaning	Why it matters
Upregulated genes	Genes higher after treatment.	Activated response programs.
Downregulated genes	Genes lower after treatment.	Suppressed programs.
Effect size or z-score	Magnitude and direction of change.	Feature strength and ranking.
Metadata	Drug, dose, time, cell line.	Essential for context and comparability.

Abbreviations in this section

Abbreviation	Expansion	Meaning
LINCS	Library of Integrated Network-based Cellular Signatures	Large perturbational profiling effort.
L1000	Landmark 1000 gene-expression assay	Reduced assay that measures landmark genes and infers others.
CMap	Connectivity Map	Perturbational resource connecting drugs, genes, and diseases by expression signatures.
KO	Knockout	Genetic disruption/removal of a gene.
KD	Knockdown	Reduction of gene expression.
OE	Overexpression	Artificial increase in gene expression.

20. Dose, time, and cell-line dependence

Drug effects are not universal. The same compound can produce different signatures at different doses, times, and cell lines. This is a central issue for synergy prediction.

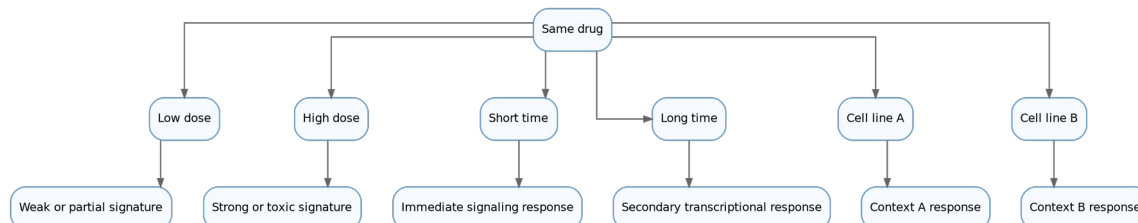


Figure 32. Perturbation signatures depend on experimental context.

Modeling consequence: A drug-pair model should condition on the cell line. A drug pair can be synergistic in one genetic/expression context and not in another.

Abbreviations in this section

Abbreviation	Expansion	Meaning
Cell line	Cultured cell population	A laboratory model with a specific genetic and expression context.
Signature	Expression-response profile	Up/down gene pattern after perturbation.

21. Recommended mental model: three biological layers

Use three layers: direct drug-target biology, network/pathway propagation, and transcriptional response. This model connects chemical compounds to expression features and synergy phenotypes.



Figure 33. Three-layer model for drug-synergy biology.

Layer	Data sources	Feature examples
Drug-target	DrugBank, ChEMBL, DGIdb, CTD.	Target identity, target class, target potency.
Network/pathway	KEGG, Reactome, STRING, BioGRID.	Pathway membership, neighborhood, propagation scores.
Transcriptional response	LINCS, L1000, CMap, RNA-seq.	Up/down genes, pathway activity, regulon activity.

Abbreviations in this section

Abbreviation	Expansion	Meaning
PPI	Protein-protein interaction	Protein association/interaction network.

22. Feature families for a synergy model

The goal is not to replace your current features, but to add biological structure. Each feature family answers a different question.

Feature family	Examples	Question answered
Drug chemical features	SMILES embeddings, fingerprints.	What does the molecule look like chemically?
Drug-target features	Known targets, target classes, potency.	What does the drug directly hit?
Target pathway features	KEGG/Reactome pathway membership.	What biological processes contain the targets?
Network-expanded features	STRING/BioGRID neighborhoods, propagation scores.	What nearby proteins may be affected?
Perturbation features	LINCS/L1000 signatures, up/down genes.	What expression response has been observed after treatment?
Cell-line features	Basal expression, pathway scores, TF scores, mutations/CNV.	What vulnerabilities exist before treatment?
Pair features	Target overlap, pathway complementarity, network proximity, signature similarity/opposition.	How do the two drugs interact biologically?

Abbreviations in this section

Abbreviation	Expansion	Meaning
SMILES	Simplified Molecular Input Line Entry System	Text representation of chemical structure.
CNV	Copy-number variation	Gain or loss of genomic DNA segments.

23. Example drug-pair construction

Suppose Drug A is a MEK inhibitor and Drug B is a PI3K inhibitor in a KRAS-mutant cancer cell line. KRAS can drive MAPK signaling, while PI3K/AKT can provide survival signaling. A pair that blocks both growth and survival may be synergistic in the right context.

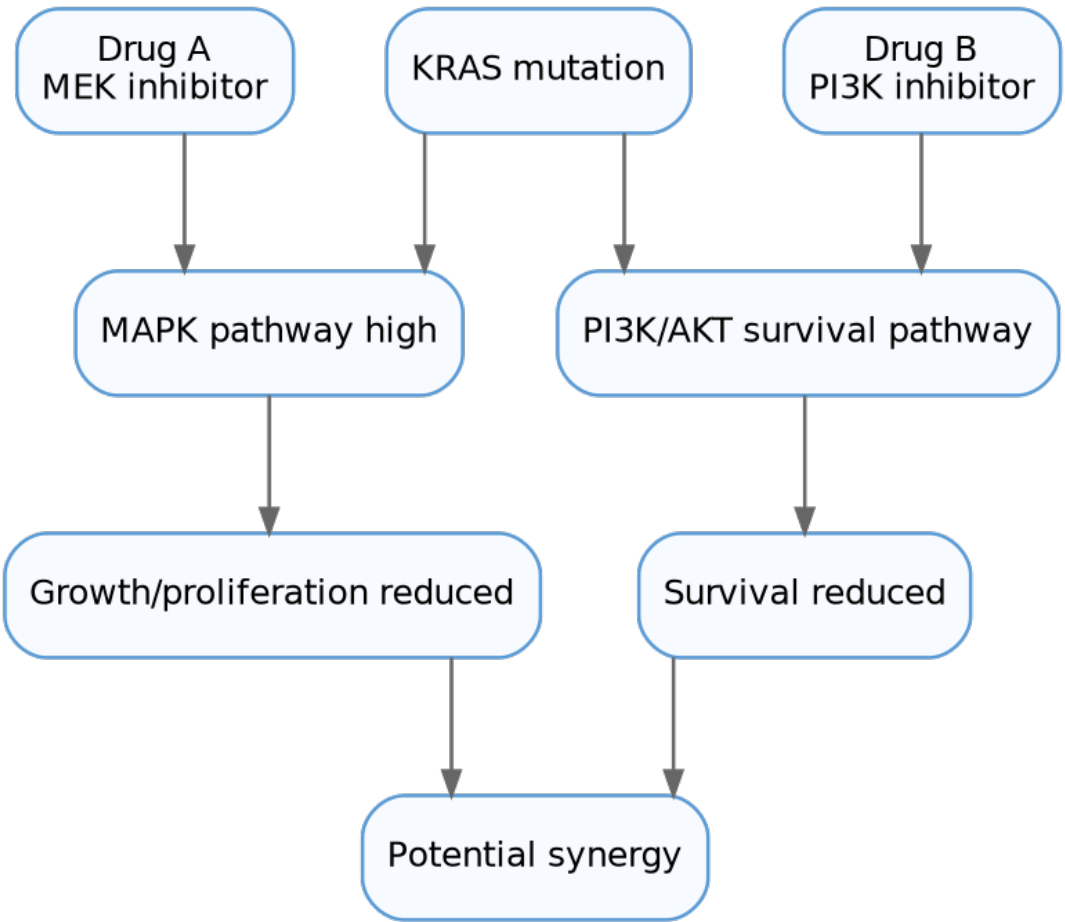


Figure 34. Example of complementary pathway blockade.

Feature	Value in example	Interpretation
Drug A target	MEK.	Growth-pathway blockade.
Drug B target	PI3K/AKT axis.	Survival-pathway blockade.
Cell context	KRAS mutation, high MAPK activity.	The cell may depend on MAPK signaling.
Pair relationship	Parallel growth + survival blockade.	Potential biological synergy.

Abbreviations in this section

Abbreviation	Expansion	Meaning
KRAS	Kirsten rat sarcoma viral oncogene homolog	A RAS-family gene often mutated in cancer.

24. Converting 23k genes into model-ready biological programs

Instead of feeding only a raw 23k-gene expression vector, you can compute compact biological programs. This reduces dimensionality, improves interpretability, and aligns features with known biology.

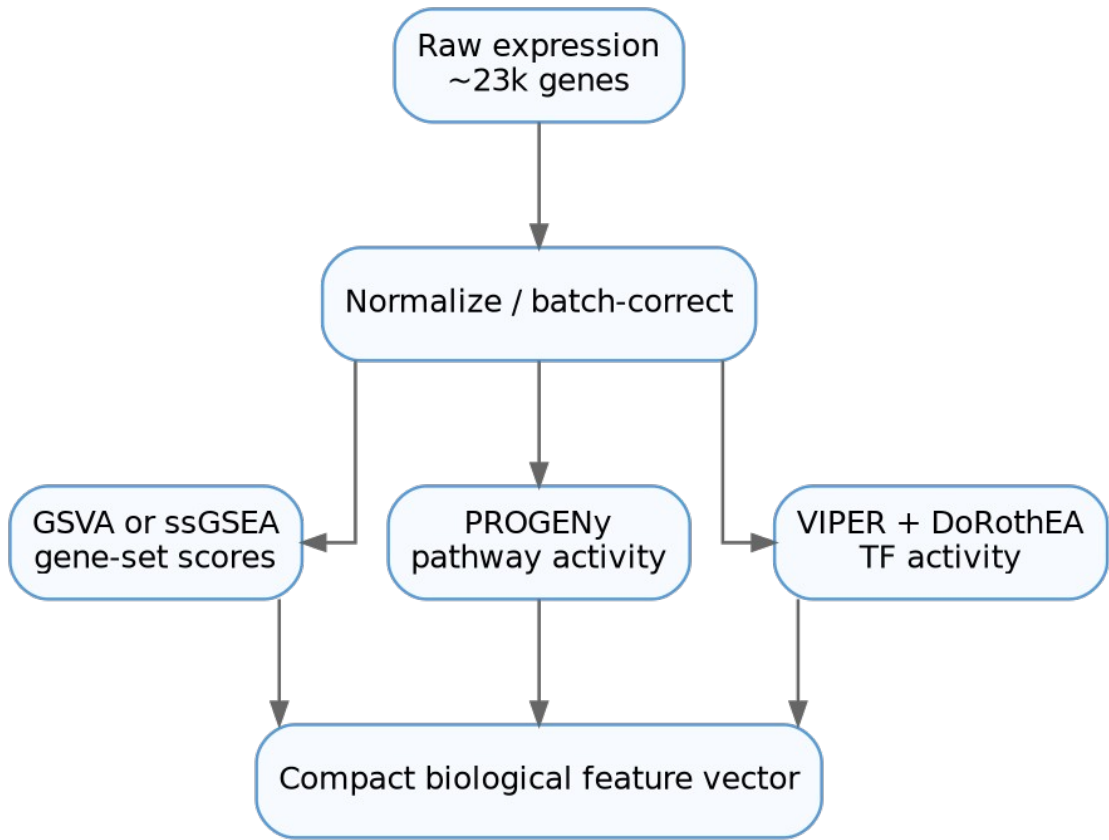


Figure 35. Converting raw expression to biological program features.

Example score	Possible interpretation
MAPK_activity = high	Growth/proliferation signaling may be active.
PI3K_activity = high	Survival/growth signaling may be active.
p53_activity = low	Stress-response function may be impaired or suppressed.
MYC_activity = high	Proliferation/transcriptional amplification may be active.
Apoptosis_score = low	The cell may resist death signals.

Abbreviations in this section

Abbreviation	Expansion	Meaning
Batch correction	Removal of technical variation	Preprocessing to reduce non-biological differences between datasets.

25. Architecture options

25.1 Strong tabular baseline

A biology-augmented tabular model is the best immediate next step because it is interpretable and easy to ablate. Combine chemical, target, pathway, perturbation, and cell-line features.

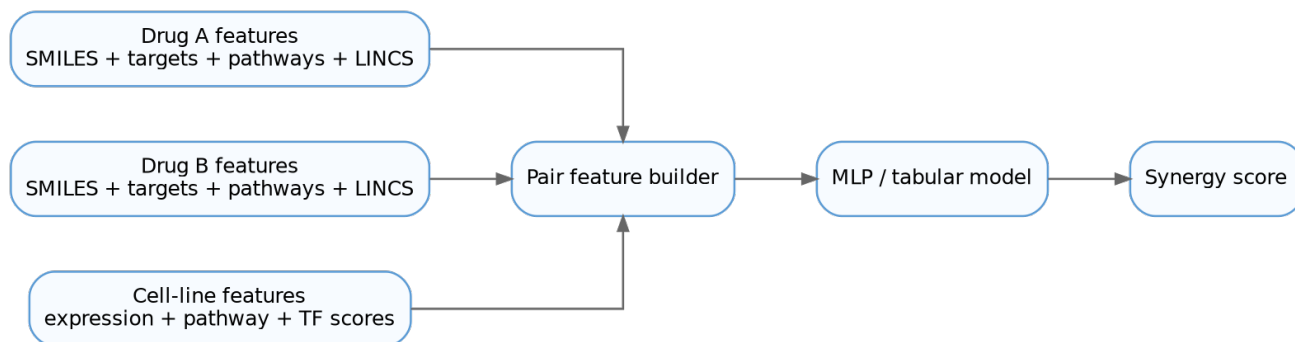


Figure 36. Biology-augmented tabular baseline.

25.2 Multimodal encoder

A multimodal encoder learns separate representations for chemistry, drug biology, and cell-line context, then fuses them for prediction. This is useful when you have enough data.

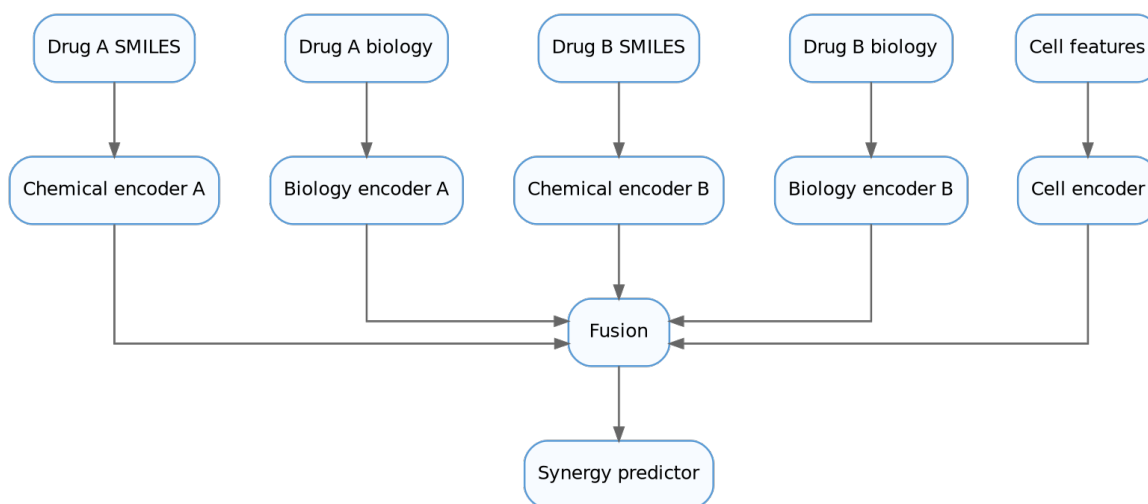


Figure 37. Multimodal synergy architecture.

25.3 Graph-aware model

A graph-aware model can use biological networks directly through propagation features or graph neural networks. This is attractive later, but only after the data pipeline and baselines are clean.

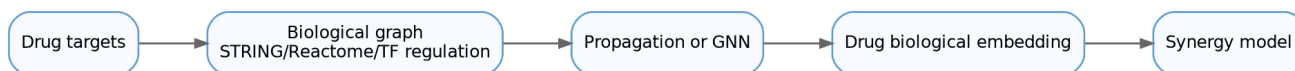


Figure 38. Graph-aware drug-biology representation.

Abbreviations in this section

Abbreviation	Expansion	Meaning
MLP	Multilayer perceptron	A feed-forward neural network.
GNN	Graph neural network	A neural network designed for graph-structured data.

26. Symmetric drug-pair fusion

Drug A + Drug B is the same combination as Drug B + Drug A. Your architecture should respect this symmetry unless the experiment has an order effect. A common approach is to fuse drug embeddings using sum, absolute difference, and elementwise product.

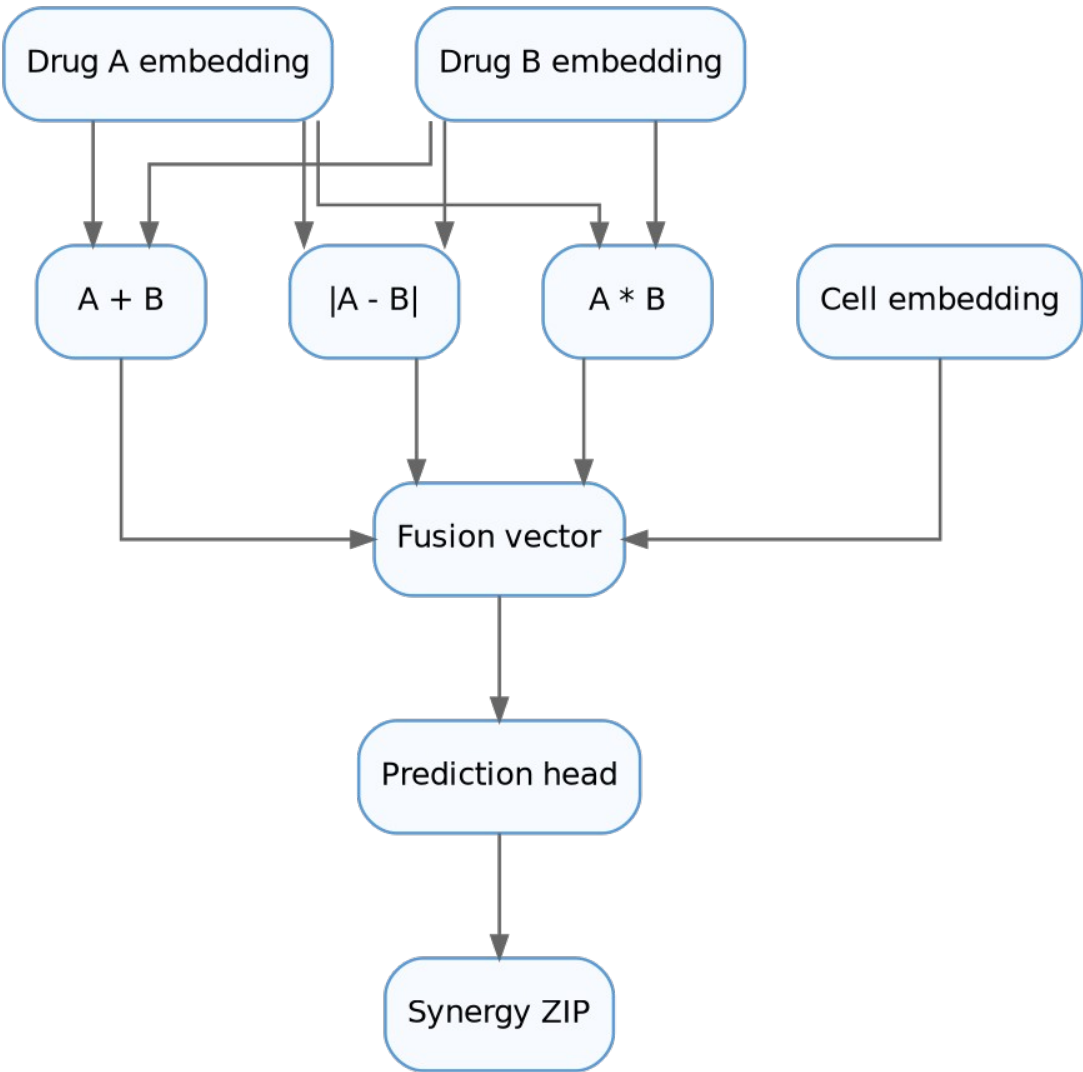


Figure 39. Symmetric pair fusion for drug-combination prediction.

Practical fusion vector: Use [A+B, abs(A-B), A*B, cell_embedding, A-cell interaction, B-cell interaction] as a strong baseline design.

Abbreviations in this section

Abbreviation	Expansion	Meaning
ZIP	Zero Interaction Potency	A drug synergy scoring framework/metric.
Embedding	Learned vector representation	A numeric vector summarizing an entity such as a drug or cell line.

27. Ablation experiments and practical research roadmap

Ablation experiments tell you whether each biological feature family actually helps. Add one feature family at a time and evaluate generalization under meaningful splits.

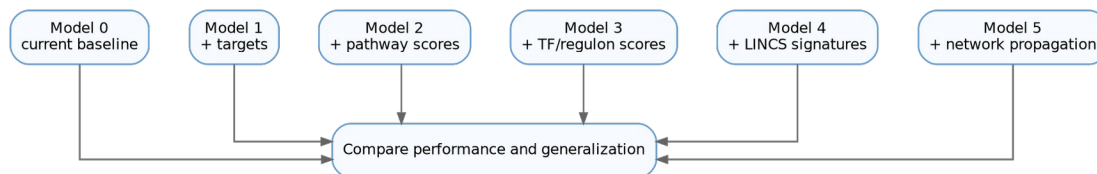


Figure 40. Ablation ladder for testing biological feature value.

Evaluation dimension	Recommended checks
Regression metrics	RMSE, MAE, Pearson correlation, Spearman correlation.
Classification/ranking metrics	AUROC and AUPRC if converting synergy into classes.
Generalization splits	Random split, leave-drug-out, leave-cell-line-out, leave-drug-pair-out.
Interpretability checks	Which pathways, targets, or regulons drive predicted synergy?

1. For each drug, get known targets from DrugBank, ChEMBL, DGIdb, and CTD where possible.
2. Map drug targets to standardized genes/proteins.
3. Expand targets through STRING, Reactome, and KEGG neighborhoods.
4. Convert affected neighborhoods into pathway and regulon scores.
5. Compare inferred biology with LINCS/L1000 perturbation signatures when available.
6. Feed these as drug-context features into the synergy model.
7. Use ablations to prove which feature families improve performance and generalization.

Final take-home: Do not rely only on KEGG or STRING to infer drug-gene effects. Use drug-target data, pathway/network expansion, regulon and pathway scoring, and perturbational transcriptomics together.

Abbreviations in this section

Abbreviation	Expansion	Meaning
RMSE	Root mean squared error	Regression error metric.
MAE	Mean absolute error	Regression error metric.
AUROC	Area under the receiver operating characteristic curve	Classification ranking metric.
AUPRC	Area under the precision-recall curve	Useful when positive labels are imbalanced.

Final integrated mental model

The complete mental model is: drug -> target protein -> protein activity change -> pathway or network change -> transcription-factor activity change -> gene-expression change -> phenotype -> synergy. Cell-line context influences almost every step.

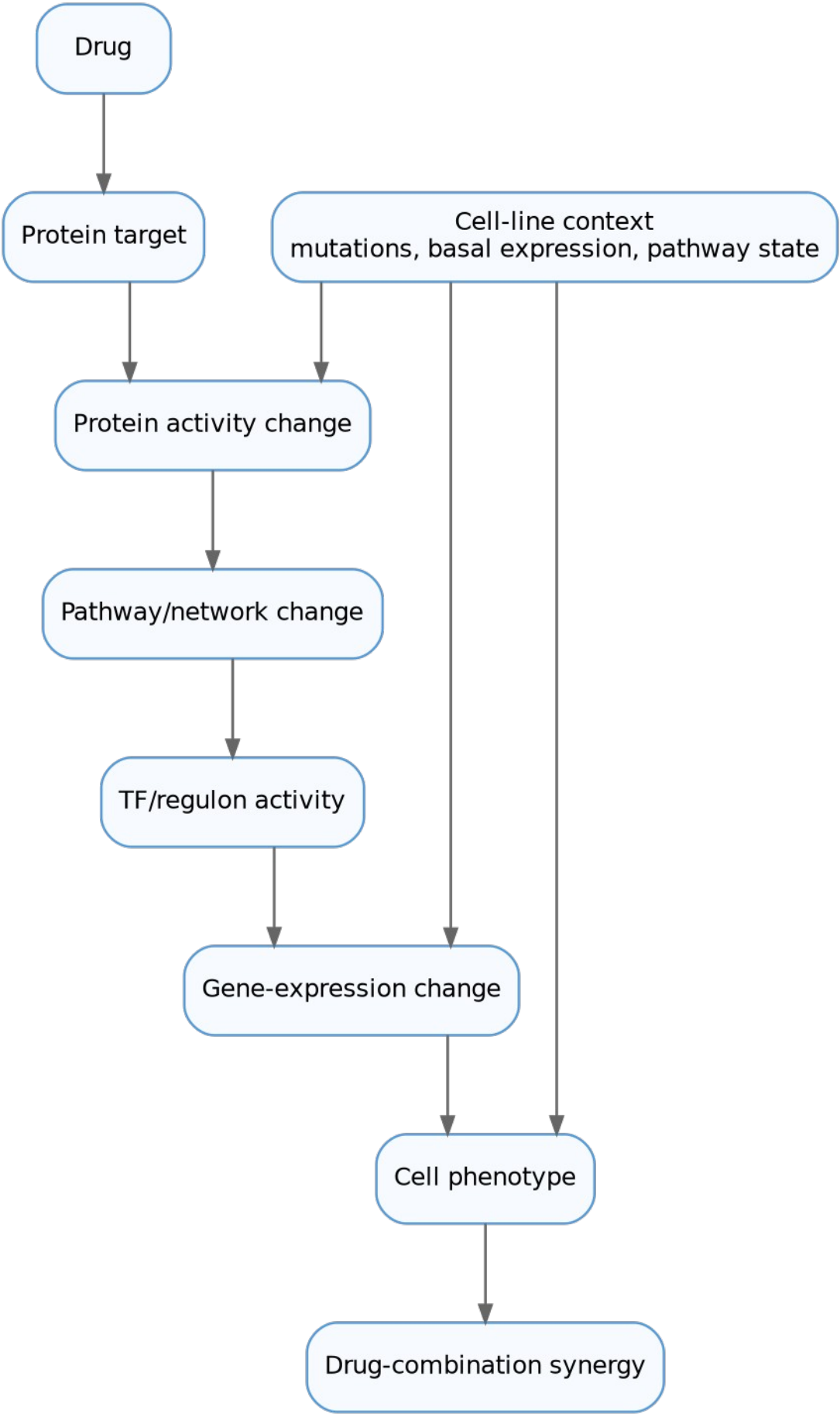


Figure 41. Integrated model of drug effect and synergy.

Research direction: Represent drugs as chemistry plus target/pathway/perturbation biology. Represent cells as basal expression plus pathway and TF activities. Represent pairs using target overlap, pathway complementarity, network proximity, and signature relationships.

Selected resources

- NCBI Bookshelf and Molecular Biology of the Cell for central dogma, transcription, translation, and gene regulation.
- Reactome and KEGG for curated pathway/reaction knowledge.
- STRING and BioGRID for protein interaction or association networks.
- DrugBank, ChEMBL, DGIdb, and CTD for drug-target or chemical-gene evidence.
- DoRothEA, TRRUST, and ChEA for transcription-factor regulatory relationships.
- GSEA, ssGSEA, GSVA, PROGENy, and VIPER for enrichment, pathway, and regulon scoring.
- LINCS L1000 and Connectivity Map for perturbational transcriptomic signatures.
- NetworkX, graph-tool, or PyTorch Geometric for computational network analysis and graph modeling.